Smart Waste Management System

3.1 Functional Requirements

Req#	Requirements	Description	Comments	Rate
FR_01	Bin Monitoring Management	The system shall monitor simulated bin fill levels, bin type, and operational status (normal, full, damaged, maintenance required).	Enables real-time visibility of bin conditions across the city.	High
FR_02	Bin Registration Management	The system shall allow authorized users to register new bins with unique ID, type, capacity, location, and assigned service zone.	Ensures accurate identification and management of bins.	High
FR_03	Bin Information Update	The system shall allow authorized users to update bin details, including location, type, capacity, and operational status.	Keeps bin records accurate and up to date.	Medium
FR_04	Bin Deactivation Management	The system shall allow authorized users to deactivate bins that are removed, replaced, or no longer in service. priority.	Prevents inactive bins from affecting operations and reports.	Medium
FR_05	Bin History Tracking	The system shall store historical data for each bin, including fill level changes, collection events, alerts, and maintenance records.	Supports performance analysis and operational review.	High
FR_06	Map Visualization	The system shall display bins, trucks, and service zones on an interactive map with visual status indicators.	Improves operational awareness for managers and operators.	High
FR_07	Map Filtering and Search	The system shall allow users to filter and search map data by zone, bin status, bin type, and truck status.	Helps users quickly locate relevant operational information.	Medium

FR_08	Overflow Alert Generation	The system shall automatically generate alerts when a bin exceeds a predefined fill threshold or reports an abnormal condition.	Supports faster intervention and prevents overflow.	High
FR_09	Alert Acknowledgment	The system shall allow authorized users to acknowledge alerts before resolution.	Improves control over alert handling and response progress.	Medium
FR_10	Alert Escalation Managemen t	The system shall escalate unresolved critical alerts to higher-level staff after a defined time period.	Ensures important incidents are not ignored.	High
FR_11	Alert Priority Classification	The system shall classify alerts by severity level such as low, medium, and high.	Supports prioritization of urgent operational issues.	High
FR_12	Collection Route Optimization	The system shall generate optimized waste collection routes based on bin fill levels, location, truck capacity, and service zone priority.	Reduces unnecessary trips, fuel use, and travel time.	High
FR_13	Dynamic Route Adjustment	The system shall allow routes to be updated dynamically when new incidents, citizen reports, or full bins appear during operation.	Maintains efficiency in changing field conditions.	High
FR_14	Manual Schedule and Route Override	The system shall allow managers to manually modify schedules and routes when necessary.	Provides flexibility in emergencies or special situations.	Medium
FR_15	Collection Scheduling	The system shall create daily and weekly collection schedules based on bin fill levels, zone demand, and fleet availability.	Supports demand-based collection planning.	High
FR_16	Route Execution Tracking	The system shall track the execution status of each assigned route as planned, in progress, completed, delayed, or skipped.	Provides visibility into route completion and delays.	High
FR_17	Missed Collection Recording	The system shall record missed collections and the reason for each missed stop.	Improves accountability and service quality analysis.	High
FR_18	Fleet Tracking	The system shall track truck status, assigned drivers, simulated live location, fuel usage, and availability.	Improves coordination and fleet control.	High

FR_19	Truck Registration Management	The system shall allow authorized users to register and update truck information including vehicle ID, capacity, type, and operational status.	Supports accurate fleet management.	High
FR_20	Vehicle Maintenance Management	The system shall record maintenance schedules, maintenance history, and truck service status.	Prevents unexpected failures and supports fleet reliability.	Medium
FR_21	Driver Assignment Management	The system shall allow managers to assign drivers to trucks and collection routes.	Ensures clear responsibility for route execution.	High
FR_22	Driver Availability Management	The system shall record driver availability, shift status, and assignment eligibility for scheduling purposes.	Supports realistic and efficient planning.	Medium
FR_23	Driver Task Management	The system shall provide drivers with assigned routes, stop sequences, and task updates through a mobile-accessible interface.	Supports field execution and reduces communication delays.	High
FR_24	Collection Stop Confirmation	The system shall allow drivers to confirm completion of each collection stop.	Provides accurate progress tracking during route execution.	High
FR_25	Field Incident Reporting	The system shall allow drivers or field operators to report incidents such as damaged bins, blocked access, or other collection problems.	Improves field communication and issue handling.	High
FR_26	Recycling Monitoring	The system shall track recycling bins separately and record recycling collection data.	Improves recycling oversight and service monitoring.	High
FR_27	Recycling Contamination Recording	The system shall record contamination incidents detected in recycling bins.	Supports recycling quality control and environmental compliance.	High
FR_28	Recycling Performance Reporting	The system shall generate reports on recycling collection trends, contamination rates, and recycling performance by zone.	Supports better recycling planning and evaluation.	High
FR_29	Citizen Issue Reporting	The system shall allow citizens to submit waste-related complaints or reports including location, description, and optional photo evidence.	Replaces phone-based reporting with digital tracking.	High

FR_30	Citizen Report Validation	The system shall validate required report information before a citizen report is submitted successfully.	Improves the completeness and quality of submitted reports.	Medium
FR_31	Report Assignment and Status Tracking	The system shall assign citizen reports to managers or field operators and track each report from submission to resolution.	Ensures accountability and faster complaint handling.	High
FR_32	Citizen Report Status Viewing	The system shall allow citizens to view the current status of their submitted reports.	Improves transparency and user trust.	Medium
FR_33	Reporting and Analytics Dashboard	The system shall provide dashboards with KPIs such as collected bins, response times, truck utilization, recycling rates, and zone performance.	Supports data-driven decisions and municipal reporting.	High
FR_34	Exportable Reports	The system shall export operational and analytical reports in formats such as PDF and CSV.	Reduces manual reporting effort and supports sharing.	Medium
FR_35	User Authentication and Role Management	The system shall manage authentication and role-based access for administrators, managers, drivers, field operators, and citizens.	Protects system functions and data access.	High

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the Smart Waste Management System. They specify how well the system performs in terms of security, performance, reliability, usability, scalability, maintainability, and compliance.

3.2.1 Product Requirements

3.2.1.1 Security Requirements

- Sensitive data shall be encrypted both at rest and in transit
- Secure authentication shall be implemented for all users (administrators, managers, drivers, operators)
- Role-Based Access Control (RBAC) shall restrict access based on user roles
- All critical system actions (logins, route updates, schedule changes, report status updates) shall be logged
- Secure backup and recovery mechanisms shall be implemented to prevent data loss
- The system shall support secure integration with IoT sensors, GPS services, and external APIs

3.2.1.2 Performance Requirements

- The system shall respond to user actions within 3 seconds under normal conditions
- Route optimization shall be completed within 5 seconds for standard operational zones
- The system shall support concurrent users without performance degradation
- Dashboards, maps, and bin status updates shall refresh in near real-time

3.2.1.3 Reliability Requirements

- The system shall ensure high availability during operational hours
- Automated backups shall be performed regularly
- The system shall support fast recovery in case of failures
- Data consistency shall be maintained across all modules (bins, routes, fleet, reports)
- Monitoring and logging mechanisms shall detect and support resolution of system errors

3.2.1.4 Usability Requirements

- The system interface shall be intuitive and user-friendly
- Drivers and field operators shall be able to use the mobile interface with minimal training
- Maps, dashboards, and reports shall present information clearly and consistently
- Citizen reporting shall require minimal steps to submit issues
- The system shall provide clear feedback messages (success, warning, error)

3.2.1.5 Scalability Requirements

- The system shall support growth in the number of bins, trucks, users, and service zones
- The database shall handle increasing volumes of operational and historical data
- The system shall support future integration of IoT sensors without major redesign
- The platform shall scale from a city-level deployment to larger municipal systems

3.2.1.6 Maintainability Requirements

- The system shall be designed using a modular architecture
- Code and system components shall be well documented
- System configuration (thresholds, zones, schedules) shall be adjustable without code changes
- The system shall support future enhancements such as AI analysis and advanced routing

3.2.2 Organizational Requirements

3.2.2.1 Training and Documentation

- User manuals shall be provided for all system users
- Technical documentation shall be available for developers and maintainers
- ECO Tirana staff shall receive training before system deployment
- Help and support features shall be available within the system

3.2.2.2 Change Management

- System updates shall follow defined procedures
- Version control shall be used for managing system changes
- Rollback mechanisms shall be available in case of failed updates
- Major system changes shall be documented and approved

3.2.2.3 Environmental Requirements

- The system shall operate on web-based infrastructure
- The platform shall support both desktop and mobile access
- Required software dependencies and technologies shall be documented
- Secure network communication shall be ensured
- The system shall support operation under intermittent connectivity conditions

3.2.3 External Requirements

3.2.3.1 Integration Requirements

- The system shall support integration with IoT bin sensors
- The system shall integrate with GPS and map services
- APIs shall be available for data exchange with external systems
- The platform shall support integration with future AI services
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3.2.3.2 Ethical Requirements

- Citizen data shall be used only for operational and analytical purposes
- Complaint handling shall be transparent and traceable
- Data access shall be restricted to authorized users
- System outputs and analytics shall be objective and unbiased

3.2.3.3 Legislative Requirements

- The system shall comply with data protection and privacy regulations
- Data storage and processing shall follow legal standards
- Operational reports shall support transparency and accountability
- Security and data handling shall align with legal and institutional policies